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Generalized Jeffreys's approximate objective Bayes factor: Model-selection consistency, finite-sample accuracy, and the Bayesian evidence in early-phase clinical trials

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Abstract. We define generalized Jeffreys's approximate objective Bayes factor (eJAB) from the p-value, sample size, and test dimension, establish conditions for model-selection consistency, and assess accuracy against Markov chain Monte Carlo computed Bayes factors. Equally Weighted Decision Error Profiles compare eJAB with test-statistic Bayes factors (TSBF; Johnson, 2005; Johnson, Pramanik and Shudde, 2023), BIC approximations, and NHST significance rules. For 2×2 and 3×3 Pearson chi-squared independence tests, eJAB generally outperforms estimated-ω TSBF outside small-sample or weak-signal regimes; the trend reverses for 5×5 tables. With true ω, eJAB generally matches or underperforms TSBF. BIC differences are small for single-parameter tests; eJAB gains at weaker effects in multiparameter tests, whereas BIC benefits at stronger effects. Against $p < .05$, eJAB gains with larger samples and stronger signals; weak signals and some small-sample settings favor NHST. We also derive a null-centered one-sided extension. An exploratory reanalysis of 16,563 findings from 1,790 early-phase ClinicalTrials.gov trials flags 461 significant findings favoring the null under eJAB. Three examples illustrate these contradictions, which warrant closer scrutiny.

Key words and phrases: Jeffreys's approximate objective Bayes factor, null-hypothesis significance testing, p-values, replicability, type I error.

1. INTRODUCTION

Concerns about the replicability of scientific findings, arising in part from the overreliance on null-hypothesis significance testing (NHST) at conventional p-value thresholds (Ioannidis, 2005), have prompted calls for more stringent standards of evidence (Benjamin et al., 2018; Johnson, 2013), more judicious interpretation of statistical significance (Wasserstein and Lazar, 2016), and alternative measures for quantifying statistical evidence (Berger and Sellke, 1987; Evans, 2016; Grünwald, 2024; Halsey, 2019). One such measure is the Bayes

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factor—the ratio of marginal likelihoods under competing hypotheses—which quantifies the relative evidence in favor of either the null or the alternative hypothesis (Kass and Raftery, 1995; Kass and Wasserman, 1995; Ly, Verhagen and Wagenmakers, 2016; Morey, Romeijn and Rouder, 2016; Raftery, 1995; Wagenmakers, 2007). The evidential behavior of p-values and Bayes factors has been studied through stochastic ordering in sample size and effect size, with implications for the Jeffreys–Lindley paradox, under which the two approaches can yield conflicting conclusions (Held and Ott, 2016, 2018; Huisman, 2023; Wagenmakers and Ly, 2023; Wei, Nathoo and Masson, 2023). Although the theoretical work clarifies how the two measures behave and when they may diverge, the practical computation of Bayes factors remains challenging. Bayes factors require integration over prior distributions, often via numerical methods such as Markov chain Monte Carlo (MCMC) with model-specific tuning (JASP, 2026; Morey and Rouder, 2026; Sinharay and Stern, 2005). Approximate Bayes factors offer a computationally convenient alternative when exact or full marginal-likelihood calculations are difficult. Such approximations can be derived from asymptotic theory, information criteria, p-values and test statistics, or summary statistics.

In an early but underappreciated contribution, Jeffreys (1936) developed an approximate Bayes factor, referred to here as JAB, for testing a point-null hypothesis against a composite alternative under a normal unit-information prior centered at the maximum likelihood estimate (MLE). The resulting expression contains the Wald statistic in the exponential term and the effective sample size in the square-root term. Through a detailed reexamination of Jeffreys's work, Wagenmakers (2022) further simplified the exponential component using a piecewise approximation to the quantile function associated with the Wald statistic in JAB. When the prior is centered at the null value, as is customary in objective Bayesian hypothesis testing, a sample-size correction term is incorporated into the exponent of JAB. A one-sided version of JAB can then be obtained by multiplying the two-sided JAB by a factor reflecting the posterior mass assigned to the theoretically predicted direction (Marsman and Wagenmakers, 2017). More recently, Wagenmakers, Bartoš and Aust (2026) showed that, for single-parameter Wald-type tests, certain forms of JAB can be interpreted as approximate e-processes. Because e-processes support sequential monitoring and optional stopping while controlling the probability of accumulating misleading evidence against the null hypothesis, this connection suggests that JAB may also possess useful approximate frequentist error-control properties in sequential settings.

Another influential approximation is based on the Bayesian information criterion (BIC). For single-parameter tests, the Bayes factor approximation is asymptotically equivalent to JAB, with the likelihood-ratio test statistic replacing the Wald statistic. A related refinement is proposed by Bickel (2025), which incorporates the same sample-size correction term in Wagenmakers (2022) by raising the likelihood ratio to a correction-dependent power, thereby reducing evidential exaggeration under the null hypothesis.

A line of research develops test-statistic Bayes factors (TSBFs) in closed form from the sampling distributions of test statistics under local alternatives, rather than treating them as approximate estimates of Bayes factors (Johnson, 2005, 2008; Yuan and Johnson, 2008). More recently, the Bayes factor function has retained this core structure as a family of TSBFs evaluated over alternative hypotheses indexed by standardized effect size, while extending the class of nonlocal priors used to define these alternatives and examining how prior shape and dispersion affect evidence accumulation, operating characteristics, and evidence aggregation across replicated studies (Datta et al., 2025a,b; Johnson, Pramanik and Shudde, 2023). For each fixed nonzero effect size, the resulting Bayes factor function is pointwise model-selection consistent.

Building on these developments, we derive an extended version of JAB, hereafter *eJAB*, using the Savage–Dickey density ratio representation under a generalized unit-information prior. This extension incorporates parameter dimension, accounts for finite-sample corrections, and facilitates applications to nonparametric testing. Our work differs from Bickel (2025) in its consideration of p-values beyond those from the likelihood ratio test and in our explicit statement of conditions for model-selection consistency. For the latter, Bickel (2025) demonstrates asymptotic equivalence to the BIC so that model-selection consistency arises under the conditions specified in Neath and Cavanaugh (2012). Our work also differs from Bartoš and Wagenmakers (2023), Rostgaard (2023), and Wagenmakers (2022) primarily in the formulation of the generalization and the development of its theoretical justification. We assess the finite-sample performance of *eJAB* across ten common parametric and nonparametric tests. The results suggest that *eJAB* is a broadly applicable and reliable approach to approximate the Bayes factor. We then apply *eJAB* to 16,563 reported primary- and secondary-outcome findings from 1,790 early phase I, phase I, and phase II trials registered on https://ClinicalTrials.gov (CTG), accessed on August 7th, 2025, and argue that NHST (p-value $\leqslant \alpha$) and $eJAB_{01} > 1$ (favoring the null) contradictions are candidate type I errors. We flag 461 such results and

report the distribution of statistical evidence separately across the three early-phase classifications for primary and secondary outcomes in CTG. We also identify 19 instances of the Jeffreys–Lindley paradox.

2. METHODOLOGY

DEFINITION 2.1. To test the hypotheses $\mathcal{H}_0 : \boldsymbol{\theta} = \boldsymbol{\theta}_0$ versus $\mathcal{H}_1 : \boldsymbol{\theta} \neq \boldsymbol{\theta}_0$, we define *eJAB* as

$$eJAB_{01} = \sqrt{n} \exp\left\{ -\frac{1}{2} \frac{n^{1/q} - 1}{n^{1/q}} Q_{\chi^2_q}(1-p) \right\},$$

where q is the dimension of the test, n is the sample size, $Q_{\chi^2_q}(\cdot)$ is the quantile function of the chi-squared distribution with q degrees of freedom (df), and p is the p-value from an NHST. We further define $eJAB_{10} = 1 \,/\, eJAB_{01}$.

For a scalar right-sided test of $\mathcal{H}_0 : \theta = \theta_0$ versus $\mathcal{H}_+ : \theta > \theta_0$, let p_1 denote the one-sided p-value and define $p = 2\min\{p_1, 1-p_1\}$. We use a symmetric local prior centered at the null value θ_0. Our finite-sample convention uses the corresponding Savage–Dickey approximation with $(n-1)/n$ omitted from the exponent. Applying the separate directional correction of Wagenmakers (2022) then gives

$$eJAB_{0+} = \frac{\sqrt{n}}{2(1-p_1)} \exp\left\{ -\frac{1}{2} Q_{\chi^2_1}(1-p) \right\}.$$

The directional correction uses prior mass $1/2$ and approximate posterior mass $1-p_1$ in the specified direction, for moderate sample sizes and a relatively vague local prior. The left-sided test is handled analogously. In contrast, the two-sided $eJAB_{01}$ retains the finite-sample correction because it is based on an MLE-centered prior. Appendix C.5 provides implementation details for the clinical-trial analysis.

THEOREM 2.2. *Assuming the NHST p-value satisfies two regularity conditions,*

- ***R1***: $p \xrightarrow{D} \mathrm{Unif}(0,1)$ *as* $n \to \infty$ *under* $\mathcal{H}_0$,
- ***R2***: $D_n = -\sqrt{n} \cdot p \ln p \xrightarrow{P} 0$ *as* $n \to \infty$ *under* $\mathcal{H}_1$,

then,

$$\operatorname*{plim}_{n\to\infty} eJAB_{01} = \begin{cases} \infty & \text{under } \mathcal{H}_0, \\ 0 & \text{under } \mathcal{H}_1. \end{cases}$$

The proof is provided in Appendix D.

For $q = 1$, write $x = Q_{\chi^2_1}(1-p)$. The two definitions satisfy

$$eJAB_{0+} = \frac{eJAB_{01} \exp\{-x/(2n)\}}{2(1-p_1)}.$$

Under $\mathcal{H}_0$, **R1** gives $x = \mathcal{O}_p(1)$, so the exponential factor converges in probability to 1; since $[2(1-p_1)]^{-1} \geqslant 1/2$, the one-sided eJAB diverges in probability. Under $\mathcal{H}_+$, the exponential factor is at most 1. Thus **R2**, together with **R3**: $(1-p_1)^{-1} = \mathcal{O}_p(1)$ under $\mathcal{H}_+$, implies that the one-sided eJAB converges in probability to zero.

FACT. We can derive *eJAB* as an approximation using a generalization of the multivariate normal unit-information prior for the parameters under test,

$$\boldsymbol{\theta} \sim \boldsymbol{\mathcal{N}}_q\Big(\hat{\boldsymbol{\theta}},\ n^{1/q} \cdot \mathbf{J}_n^{-1}(\hat{\boldsymbol{\theta}})\Big),$$

where $\hat{\boldsymbol{\theta}}$ is the MLE, and $\mathbf{J}_n(\hat{\boldsymbol{\theta}})$ is the observed information matrix of size $q \times q$. Under an asymptotic Gaussian approximation, the posterior density takes the form, $\pi(\boldsymbol{\theta} \mid \boldsymbol{y}, \mathcal{H}_1) \approx$

$$(2\pi)^{-\frac{q}{2}} |\mathbf{J}_n(\hat{\boldsymbol{\theta}})|^{\frac{1}{2}} \exp\left\{ -\frac{1}{2} (\boldsymbol{\theta} - \hat{\boldsymbol{\theta}})^\top \cdot \mathbf{J}_n(\hat{\boldsymbol{\theta}}) \cdot (\boldsymbol{\theta} - \hat{\boldsymbol{\theta}}) \right\}.$$

Assuming that the prior density for any nuisance parameters under $\mathcal{H}_0$ is everywhere equal to the conditional prior density for those parameters under $\mathcal{H}_1$, given the null value of the parameter under test, the Bayes factor (BF) can be expressed as the Savage–Dickey density ratio (Mulder, Wagenmakers and Marsman, 2022),

$$BF_{01} = \frac{\pi(\boldsymbol{\theta}_0 \mid \boldsymbol{y}, \mathcal{H}_1)}{\pi(\boldsymbol{\theta}_0 \mid \mathcal{H}_1)} \approx \sqrt{n} \exp\left\{ -\frac{1}{2} \frac{n^{1/q} - 1}{n^{1/q}} W \right\},$$

where $\boldsymbol{y}$ represents the data, and W is the Wald statistic. When the p-value is obtained from a Wald test, substituting $W = Q_{\chi^2_q}(1-p)$ yields $eJAB_{01}$ in Definition 2.1.

The prior underlying the approximate BF interpretation is centered around the MLE $\hat{\boldsymbol{\theta}}$ with $|\mathrm{Var}(\boldsymbol{\theta})| = n\,|\mathbf{J}_n^{-1}(\hat{\boldsymbol{\theta}})|$. When $q = 1$, this is the unit-information prior. More generally, the volume of prior probability ellipsoids around the MLE is $\mathcal{O}(n^{(1-q)/2})$. The prior distribution is asymptotically diffuse relative to the likelihood.

FACT. When the p-value is obtained from a likelihood ratio test and $q = 1$, *eJAB* reduces to the evidential BIC method of Bickel (2025). This approximation is derived by applying a finite-sample correction to the BIC approximation to the BF, which amounts to raising the likelihood ratio to the power $(n-1)/n$. Model-selection consistency for the evidential BIC follows from its asymptotic equivalence to BIC together with standard consistency results for BIC (Bickel, 2025; Neath and Cavanaugh, 2012). By contrast, the conditions of Theorem 2.2 are broadly applicable: they cover nonparametric tests, provided **R1** and **R2** hold.

3. SIMULATION STUDIES

We conduct 12 simulation studies across ten parametric and nonparametric tests to evaluate the regularity conditions and finite-sample accuracy. For each design, 5,000 datasets are generated under the null hypothesis and the alternative hypothesis with small, medium, and large effect sizes across various sample sizes. The designs include:

1. Two-sample t-test;
2. Simple linear regression;
3. Simple logistic regression;
4. One-way analysis of variance (ANOVA);
5. One-way repeated-measures ANOVA (rANOVA) with high and low intraclass correlations $\rho \in \{.9, .2\}$;
6. Chi-squared tests for independence under multinomial (total count fixed) and product-multinomial (row counts fixed) designs;
7. Cox proportional-hazards regression with right censoring;
8. Wilcoxon signed-rank test;
9. Mann–Whitney U-test;
10. Kruskal–Wallis H-test.

For nonparametric tests (items 8 through 10), we simulated using right-skewed and heavy-tailed (non-Gaussian) distributions. Detailed simulation settings, additional designs (e.g., two-way ANOVA and large contingency tables), and the `R` code for the subsequent subsections are available in the GitHub repository (see files within 'Theorem', 'Simulations' [parametric], and 'Supplementary Information' [nonparametric]).

We used p-values from the t-test for items 1 and 2; the F-test for items 4 and 5; the Wald test for items 3 and 7; and the asymptotic null distribution for item 6. The p-values for 8–10 are based on asymptotic distributions. The effective sample size n was taken as the number of uncensored observations for survival analysis (Volinsky and Raftery, 2000) and as the number of independent observations (i.e., the product of the number of participants and one less than the number of conditions) for rANOVA (Masson, 2011; Nathoo and Masson, 2016). For the other designs, n always refers to the total number of observations or counts. Among the tests with multiple df ($q > 1$) are items 4 through 6, and 10.

3.1 Verifying the Regularity Conditions

Condition **R1** of Theorem 2.2 is satisfied whenever the p-value is computed from an exact or asymptotic null distribution with a continuous cumulative distribution function.

It is not difficult to verify that **R2** is insufficient to ensure model-selection consistency of the BIC approximation to the BF,

$$BF_{01}^{(\mathrm{BIC})} = n^{\frac{q}{2}} \exp\left\{-\frac{1}{2} Q_{\chi^2_q}(1-p)\right\}.$$

Using the asymptotic expansion of the incomplete gamma function (e.g., Daouia, Stupfler and Usseglio-Carleve, 2025, Lemma A.2), $\frac{1}{2} Q_{\chi^2_q}(1-p) = -\ln p + (\frac{q}{2} - 1)\ln(-\ln p) - \ln\Gamma(\frac{q}{2}) + o(1)$. It follows that $BF_{01}^{(\mathrm{BIC})} \approx \Gamma(\frac{q}{2})\, n^{\frac{q}{2}} \cdot p\,(-\ln p)^{1-\frac{q}{2}}$, which requires a substantially stronger condition than **R2** for model-selection consistency when $q > 1$.

3.2 Accuracy of the Bayes-Factor Approximation

To assess the finite-sample accuracy of an approximation to the BF, we compared *eJAB* with alternative computations in seven studies based on the parametric designs, using fixed representative sample sizes. In this subsection, item 7 represents a parametric survival regression model with an exponential baseline and expected censoring fractions of 23% under the null, and 21%, 19%, and 18% under increasing effect sizes. The MCMC-based BFs were computed by sampling the marginal posterior density using '*rstan*' (Stan, 2025), applying logspline density estimation (Kooperberg and Stone, 1992), and calculating the Savage–Dickey density ratio. In these cases, the model underlying the MCMC sampler assumed the same prior as *eJAB*, so discrepancies reflect errors of approximation rather than differences in prior.

Pairs of ($\ln eJAB_{10}$, $\ln BF_{10}$) are color- and shape-coded by the p-value. Overall, *eJAB* tracks the MCMC-based BFs closely in Figure 1A. Some inaccuracy, $\ln eJAB_{10} < \ln BF_{10}$, appears for logistic regression when evidence favors $\mathcal{H}_1$. Replacing *eJAB* with the BIC approximation to the BF performs well when $q = 1$, but marked severe discrepancies for $q > 1$, indicating that *eJAB* better approximates BF for ANOVA and rANOVA in Figure 1B.

In sum, *eJAB* provides an accurate approximation to Bayes factors across a range of designs and effect sizes for parametric tests.

3.3 Bayesian Hypothesis Tests Using Nonparametric Statistics

Classical nonparametric tests do not specify a sampling distribution for the data, so marginal likelihoods are not defined. As such, the Bayes factor can be defined based on the marginal likelihood of the test statistic rather than on the data (Yuan and Johnson, 2008). We compared *eJAB* with Bayes factors constructed from nonparametric statistics (*BFNP*) across items 8 through 10. Appendix B reports sequences of box plots

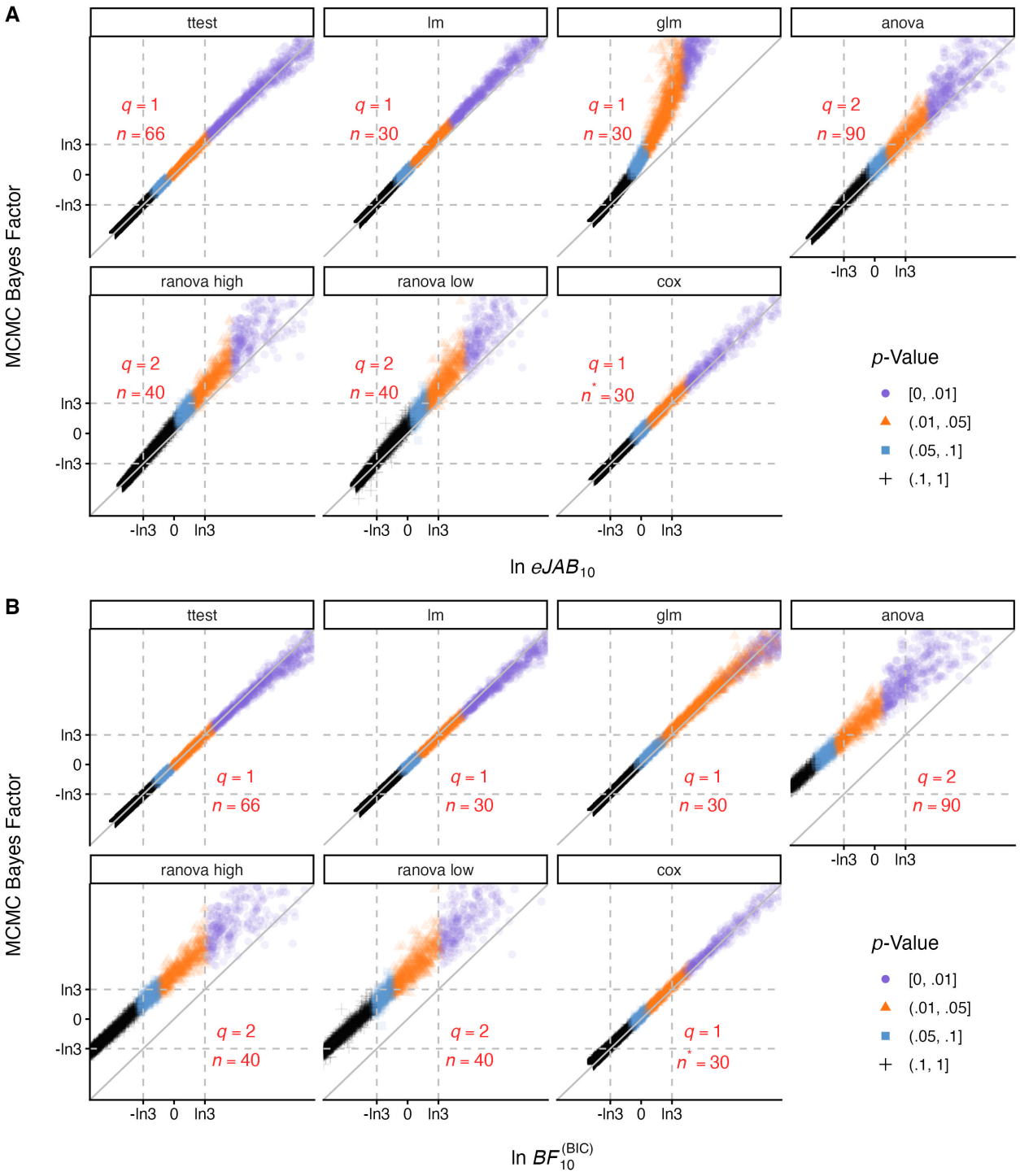


FIG 1. Comparison of the natural logarithms of (A) $eJAB_{10}$ and (B) the BIC approximation with the MCMC-based Bayes factor. Red text indicates the test dimension q and effective sample size n, except for the Cox model, for which n^* denotes the number of observations. Dashed lines indicate Bayes factors of 1/3 and 3.

depicting how the sampling distributions of *eJAB* and *BFNP* vary with effect size and sample size. Figures depicting these results, a detailed description of the experiments, and the associated `R` Markdown files are available in the GitHub repository (see the 'Supplementary Information' directory).

Under $\mathcal{H}_1$, when effects and samples are both relatively small, *BFNP* tends to provide stronger evidence for $\mathcal{H}_1$ than *eJAB*. With larger effects or larger samples, the two methods produce comparable results. Under $\mathcal{H}_0$, *eJAB* effectively quantifies evidence for $\mathcal{H}_0$.

3.4 Equally Weighted Decision Error Profiles

For each effect size θ and sample size n, we compare a simple null (effect zero) with a simple alternative (effect θ). Under 0–1 loss, the equally weighted decision error (EWDE) for decision rule d is

$$\mathrm{EWDE}_d(\theta, n) = \frac{\alpha_d(\theta, n) + \beta_d(\theta, n)}{2},$$

where α_d is the type I error probability under the null and β_d is the type II error probability at the specified alternative, equal to one minus power. Define $\Delta E = \mathrm{EWDE}_{\mathrm{comparison}} - \mathrm{EWDE}_{\mathrm{eJAB}}$. The profiles display the normalized EWDE comparison,

$$\mathrm{EWDE}_{\mathrm{norm}}(\theta, n) = \frac{\Delta E(\theta, n)}{M},$$

where $M = \max_{(\theta,n)\in\mathcal{G}} |\Delta E(\theta, n)|$ and $\mathcal{G}$ is the displayed grid for each panel. Positive values favor eJAB; negative values favor the comparison rule. Figure 2 compares eJAB with estimated- and true-effect TSBF rules (A–B), the $p < .05$ rule (C), and the BIC Bayes-factor approximation (D). For Panel A, $\hat{\omega} = \sqrt{X^2/n}$ and $\hat{\tau}^2 = X^2/q$, where X^2 is the observed chi-squared statistic; Panel B uses the true standardized effect ω with $\tau_\omega^2 = n\omega^2/q$. Panel D uses $BF_{01}^{\mathrm{BIC}} = n^{q/2}\exp(-\Lambda/2)$, where Λ is the fitted likelihood-ratio statistic, across the six parametric-likelihood tests with available BIC comparisons. Cox regression and the three rank tests are excluded from this panel.

For Pearson chi-squared tests of independence in 2×2 and 3×3 tables, eJAB generally has lower EWDE than estimated-ω TSBF outside small-sample or weak-signal regimes. Small 3×3 samples can favor TSBF even at larger effects. The broad pattern reverses for 5×5 tables: estimated-effect TSBF generally has lower EWDE, with eJAB gains concentrated in a band of weaker effects. With true ω, eJAB generally matches or underperforms TSBF; differences can be appreciable near weak signals. BIC comparisons show small absolute differences for the single-parameter tests. For ANOVA, repeated-measures ANOVA and chi-squared independence tests, eJAB's largest gains occur in weaker-effect bands that move toward zero as sample size increases. At stronger effects, BIC can have lower EWDE because both rules have high power and BIC has lower type I error; the differences diminish at larger samples. Against the NHST significance rule $p < .05$, eJAB generally gains with larger samples and sufficiently strong signals. NHST can have lower EWDE in weak-signal bands and, for multiparameter tests, at stronger effects in small samples.

Dashed curves of the form $\theta = c/\sqrt{n}$ hold $\sqrt{n}\theta$ constant. They preserve the alternative distribution when it depends on n and θ only through that quantity, but need not hold normalized EWDE constant because the eJAB threshold also depends on n. We use a Bayes-factor threshold of 1. For Bayes factors defined by fixed priors, this threshold minimizes prior-predictive error under 0–1 loss and equal prior model probabilities; it need not minimize EWDE at each fixed effect size. The estimated-effect TSBF is a plug-in sensitivity analysis.

4. BAYES-FREQUENTIST CONTRADICTIONS IN EARLY-PHASE CLINICAL TRIALS WITH EJAB

Having evaluated $eJAB$ across a wide range of tests, we apply it to reevaluate the evidence in a large corpus of CTG results. Prior work has raised concerns about the strength of statistical evidence in clinical research. For example, Begley and Ellis (2012) reported replication for only 11% of landmark preclinical cancer findings. Hwang et al. (2016) found that most late-stage trials fail, largely for lack of efficacy. And, Sun et al. (2022) estimated that 90% of drug-development trials fail for similar reasons. We posit that failures at later phases may reflect type I errors, where the null hypothesis was incorrectly rejected at an earlier phase.

Our objective is to characterize the overall strength of evidence in 16,563 primary- and secondary-outcome CTG findings from 1,790 early-phase trials and to identify potential type I errors. These results span 36 MeSH categories (e.g., neoplasms, cardiovascular diseases, and autoimmune diseases) and encompass eight statistical tests: one-sample t-test, two-sample t-test, linear regression, logistic regression, ANOVA, Wilcoxon signed-rank test, Mann–Whitney test, and Kruskal–Wallis test.

For each, we extracted the p-value, sample size, and parameter dimension, and computed the applicable eJAB. For the 427 explicitly one-sided scalar analyses, we used $eJAB_{0+}$ without the finite-sample factor $(n-1)/n$, retaining the Wagenmakers correction $1/\{2(1-p_1)\}$; for the remaining 16,136 two-sided/unrestricted analyses, we used $eJAB_{01}$ with its finite-sample correction. Statistical significance was defined as $p_1 < .025$ and $p < .05$, respectively, retaining reported inequalities as described in Appendix C.5. We excluded survival analyses and chi-squared tests because CTG usually omits the number of uncensored observations and full contingency-table dimensions. Preprocessing to address misreported and left-censored p-values, sample sizes, and test labels (Miron, Gonçalves and Musen, 2020) is described in Appendix A.

The relationship between α and the share of reported findings with $p \leqslant \alpha$ and $eJAB_{01} > 1$ is strongly linear for $\alpha \in (0, .25)$ across the three early-phase classifications, with a visible jump at $\alpha = .05$ due to results

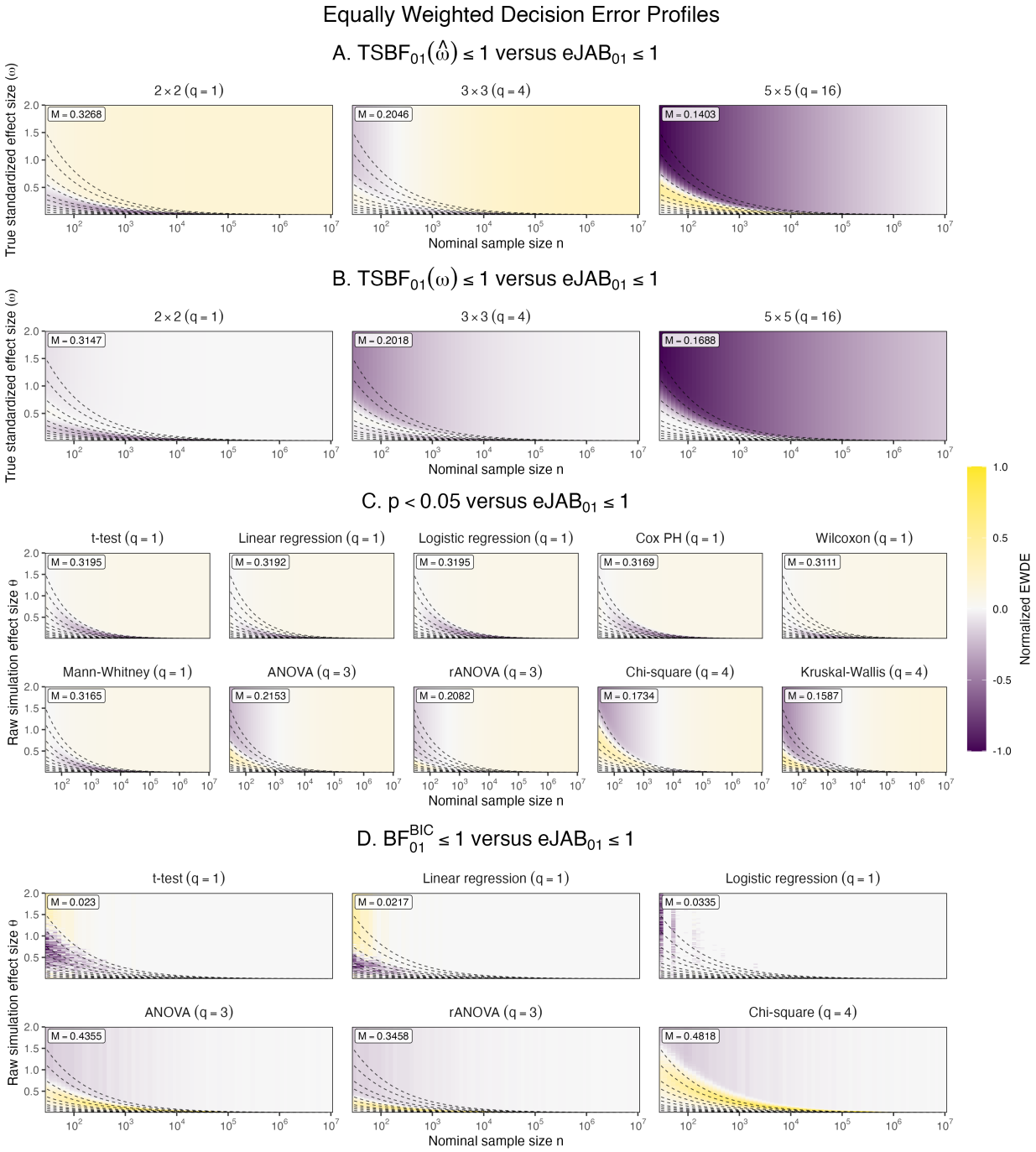


FIG 2. Equally Weighted Decision Error Profiles. (A) $TSBF_{01}(\hat{\omega}) \leqslant 1$ versus $eJAB_{01} \leqslant 1$, with $\hat{\omega} = \sqrt{X^2/n}$, and (B) $TSBF_{01}(\omega) \leqslant 1$ versus $eJAB_{01} \leqslant 1$, for 2×2, 3×3, and 5×5 contingency tables. (C) $p < .05$ versus $eJAB_{01} \leqslant 1$ across ten statistical tests. (D) $BF_{01}^{BIC} \leqslant 1$ versus $eJAB_{01} \leqslant 1$ across six parametric-likelihood tests. Colors indicate normalized EWDE: positive values (yellow) favor eJAB, whereas negative values (purple) favor the comparison rule. Each panel reports its normalization constant M, the maximum absolute EWDE difference over its displayed grid. Dashed curves represent local alternatives of the form $\theta = c/\sqrt{n}$, with $\theta = \omega$ for the contingency-table comparisons.

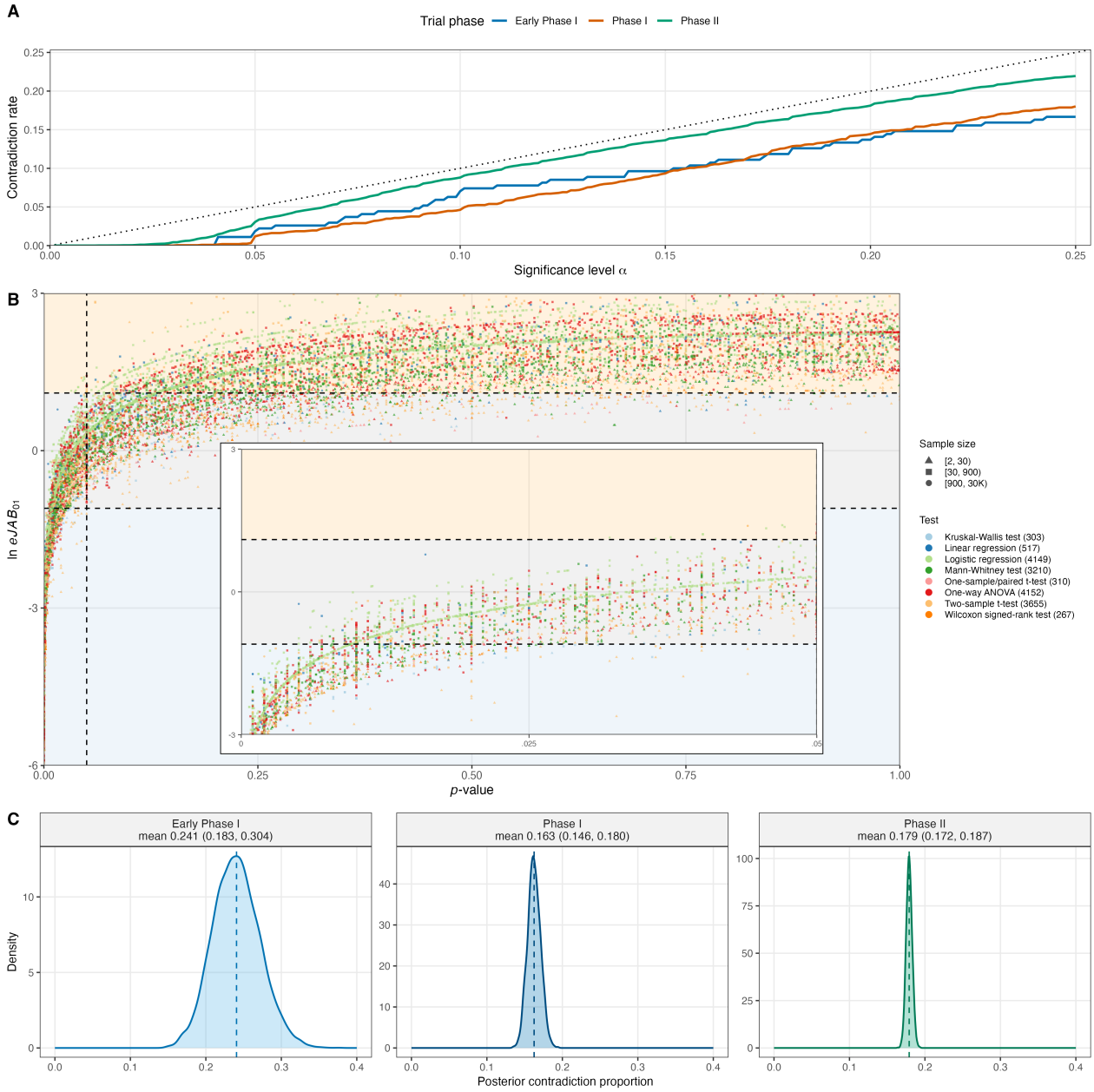


Fig 3. Clinical-trial application summaries for primary and secondary outcomes in exact Early Phase I, Phase I, and Phase II trials. (A) Contradiction rate versus significance level, with one line for each trial phase; the dotted line marks equality with α. (B) $\ln eJAB_{01}$ versus p, with colors indicating statistical tests and point shapes indicating sample-size groups. Dashed lines mark $p = .05$ and $eJAB_{01} = 1/3$ and 3; the inset enlarges the region defined by $p \leqslant .05$ and $-3 \leqslant \ln eJAB_{01} \leqslant 3$. (C) Study-accounted posterior distributions of the contradiction proportion among findings with $p \leqslant .05$, shown in separate panels for the three trial phases; dashed lines mark posterior means.

reported exactly as $p = .05$ in Figure 3A. We attribute this to left-censoring at the target threshold.

We therefore classify findings with $p \leqslant \alpha$ and $eJAB_{01} > 1$ as candidate type I errors at the significance level α. Among 6,409 results significant at $\alpha = .05$, we catalog 461 candidates. Although lower thresholds have been recommended in the literature (e.g., Benjamin et al., 2018; Johnson, 2013), $\alpha = .05$ remains germane in CTG: in a random sample of 52 trial protocols, 75% (SE 6%) explicitly set $\alpha \geqslant .05$ as the target evidential threshold. We provide the full catalog of 461 candidates in the GitHub repository, along with details for three illustrative cases cross-referenced with CTG.

We fit a hierarchical Dirichlet-multinomial model with a flat prior on study-specific proportions of $p \leqslant .05$ and $eJAB_{01} > 1/3$, aggregating draws to estimate the overall proportion. We estimate that 34.7% (posterior

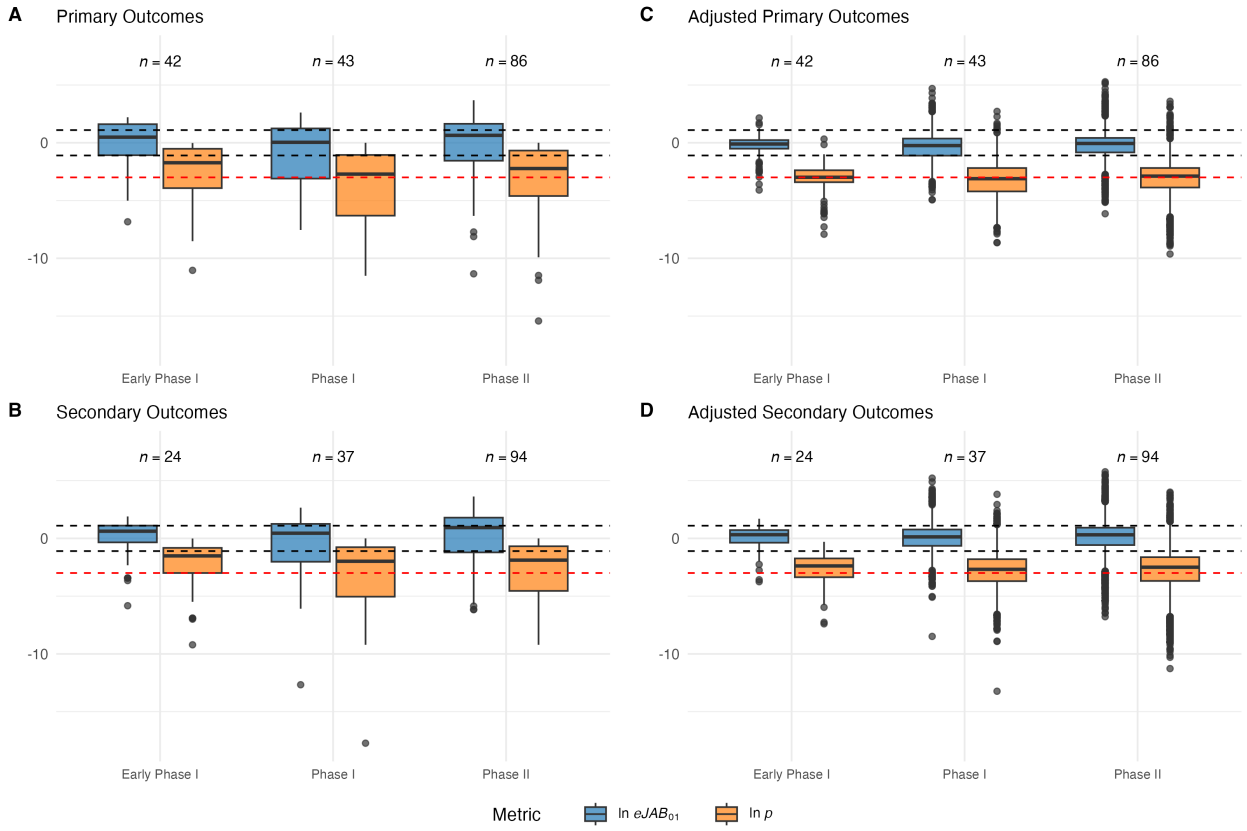


FIG 4. Measures of evidence across early phase I, phase I, and phase II clinical trials. Panels A and B show the unadjusted distributions of $\ln eJAB_{01}$ *(blue) and* $\ln p$ *(orange) for primary and secondary outcomes, respectively; panels C and D show the corresponding distributions after adjustment using linear mixed models. All panels share the same y-axis scale. The average analysis sample size is shown above each phase group. Black dashed lines mark* $-\ln 3$ *and* $\ln 3$*, and the red dashed line marks* $\ln 0.05$*.*

SD 0.46%) of $p \leqslant .05$ results correspond to $eJAB_{01} > 1/3$, i.e., evidence for the alternative no stronger than anecdotal in the early-phase analysis.

In the plot of $\ln eJAB_{01}$ versus p, with points colored by test type and shaped by sample size, the region to the left of $p = .05$ and above $eJAB_{01} = 3$ indicates instances of the Jeffreys–Lindley paradox—that is, cases where the p-value leads to rejection of $\mathcal{H}_0$, while the Bayes factor favors $\mathcal{H}_0$ over $\mathcal{H}_1$. Notably, $\ln eJAB_{01}$ shows substantial variability at the smallest p-values in Figure 3B. Contradictions with $p \leqslant .05$ yet $eJAB_{01} > 3$ occur predominantly at larger sample sizes, where observed effect sizes can be negligible in Figure 3B.

We also examine the proportion of all $p \leqslant .05$ and $eJAB_{01} \geqslant K$ for $K \in (0, 300)$; at $K = 1$, this proportion recovers the candidate type I error at $\alpha = .05$. The inset in Figure 3B highlights Jeffreys–Lindley paradox cases. We identify 19 such instances in the early-phase CTG data. Additional results stratified by MeSH category appear in Appendix B (Figures S4–S6).

In sum, the near-perfect linearity indicates that findings with $p \leqslant \alpha$ and $eJAB_{01} > 1$ behave as type I errors in the CTG data when tested at $\alpha \in (0, .2]$. To elucidate how such candidates arise, consider a Wald test with $q = 1$ and independent and identically distributed data for fixed $n < \infty$ and $\alpha > 1 - F_{\chi^2_1}\left(\frac{n \ln n}{n-1}\right)$. A candidate type I error occurs when the observed effect size falls within the interval,

$$\sqrt{\frac{1}{n} Q_{\chi^2_1}(1-\alpha)} \cdot I_1^{-\frac{1}{2}}(\hat{\theta}) \leqslant |\hat{\theta} - \theta_0| < \sqrt{\frac{\ln n}{n-1}} \cdot I_1^{-\frac{1}{2}}(\hat{\theta}),$$

where I_1 is the Fisher information for one observation. In this regime, the observed effect size is large enough to achieve standard statistical significance at sample size n. Small enough, $eJAB_{01}$ favors the null. To favor the alternative, the latter requires the observed effect size to exceed a bound of order $\mathcal{O}_p\left(\sqrt{\frac{\ln n}{n}}\right)$.

We examine the distributions of $\ln eJAB_{01}$ and $\ln p$ by trial phase for primary and secondary outcomes in Figure 4A–B. For primary outcomes, there is no discernible trend toward stronger evidence across early phase I, phase I, and phase II. For secondary outcomes, we again observe no phase-wise increase in evidence. Plotting p and sample size by phase shows that changes in p track changes in sample size (Appendix B, Figure S8).

TABLE 1

ClinicalTrials.gov-reported effects for three illustrative candidate type I errors, with Early Phase I, Phase I, and Phase II represented separately. CPR denotes cardiopulmonary resuscitation; MD, mean difference; SD, standard deviation; and SE, standard error.

CTG study	Phase	Outcome and comparison	Observed results	Method	n	p	$eJAB_{01}$	Reported effect (95% CI)
NCT00494806	Early Phase I	First flatus: rocking vs standard care	3.16 (SD .86; 34) vs 3.88 (SD .80; 32)	t-test	66	$< .05$	1.23	MD .7279 (.3174, 1.1383)
NCT03907683	Phase I	Activity after weekday move-more message	.16 (SD .69; 80)	t-test	80	.037	1.04	MD .1641 (.0098, .3185)
NCT00539539	Phase II	CPR fraction: feedback on vs off	66 (SE 1; 815) vs 64 (SE 1; 771)	Linear	1,586	.016	2.19	Adjusted MD 1.9 (.4, 3.4)

To better visualize trends in evidence across phases, we fit a linear mixed model with a fixed intercept and trial-specific random effects to each of $\ln eJAB_{01}$ and $\ln p$, and then examine residuals re-centered by the estimated intercept (Appendix B, Figure S7). The adjusted measures are essentially uniform across phases in Figure 4C–D. For primary outcomes, most adjusted p-values lie near .05 and above .01. Using $eJAB$, this interpretation corresponds to 'anecdotal evidence' or 'evidence that is barely worth mentioning' for both primary and secondary outcomes (Lee and Wagenmakers, 2014, p. 105). Comparing primary with secondary outcomes, we note a slight and phase-invariant shift toward greater evidence for the alternative in primary outcomes.

5. CONCLUSIONS

$eJAB$ can be interpreted as an approximate objective Bayes factor when p-values are obtained from Wald or likelihood-ratio tests. The same interpretation should hold for tests whose statistics differ from the Wald statistic by $o_p(1)$. Simulations suggest that this interpretation holds more broadly. In addition, $eJAB$ remains model-selection consistent for p-values from any test satisfying **R1** and **R2** of Theorem 2.2.

We compared the finite-sample performance of $eJAB$ with Bayes factors computed from MCMC and the Savage–Dickey density ratio for the t-test, linear regression, logistic regression, ANOVA, rANOVA, and parametric survival analysis. Under the same generalization of the unit-information prior, $eJAB$ closely approximates the MCMC-based Bayes factor. For a chi-squared test under multinomial or product-multinomial sampling, it agrees with the test-statistic Bayes factor under the alternative, unlike the latter, also registers evidence for the null when true. For nonparametric tests, namely Wilcoxon signed-rank, Mann–Whitney, and Kruskal–Wallis, $eJAB$ performs comparably to Bayes factors based on nonparametric test statistics. Our simulations also show improved performance compared to the BIC approximation to the Bayes factor when $q > 1$.

The Equally Weighted Decision Error Profiles compare eJAB with true- and estimated-effect TSBFs, BIC approximations, and NHST significance rules. Relative performance varies with sample size, effect size, and dimension; the comparisons do not establish uniform decision-error dominance.

In our reevaluation of 16,563 primary- and secondary-outcome findings from 1,790 early-phase clinical trials, findings with $p \leqslant \alpha$ and $eJAB_{01} > 1$ exhibit characteristics of type I errors in the CTG data when tested at significance level $\alpha \in (0, .2]$. The uniform strength of evidence across phases is consistent with a roughly constant proportion of hypotheses with true effects. The 461 CTG candidate type I errors warrant targeted follow-up to investigate replicability.

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SUPPLEMENTARY MATERIAL

Supplementary material and GitHub repository

Appendices A–D are provided in the Supplementary Material. Additional materials, referred to as the Online Appendices, including proofs, preprocessing code, extensive simulations—including nonparametric tests—CSV files containing candidate type I errors, and

code for reproducing the figures in this paper are available in the GitHub repository at https://github.com/nathoogroup/ejabclinicaltrials. An *eJAB* calculator is available at https://flxp0x-puneet-velidi.shinyapps.io/bayesfactor/.

REFERENCES

Bartoš, F. and Wagenmakers, E.-J. (2023). A general approximation to nested Bayes factors with informed priors. *Stat* **12** e600. https://doi.org/10.1002/sta4.600

Begley, C. G. and Ellis, L. M. (2012). Raise standards for preclinical cancer research. *Nature* **483** 531–533. http://dx.doi.org/10.1038/483531a

Benjamin, D. J., Berger, J. O., Johannesson, M., Nosek, B. A., Wagenmakers, E.-J., Berk, R., Bollen, K. A., Brembs, B., Brown, L., Camerer, C. et al. (2018). Redefine statistical significance. *Nature Human Behaviour* **2** 6–10. https://doi.org/10.1038/s41562-017-0189-z

Berger, J. O. and Sellke, T. (1987). Testing a point null hypothesis: The irreconcilability of p values and evidence. *Journal of the American Statistical Association* **82** 112–122. https://doi.org/10.1080/01621459.1987.10478397

Bickel, D. R. (2025). A small-sample Bayesian information criterion that does not overstate the evidence, with an application to calibrating p-values from likelihood-ratio tests. *Statistical Papers* **66** 1–17. https://doi.org/10.1007/s00362-025-01682-1

Daouia, A., Stupfler, G. and Usseglio-Carleve, A. (2025). Corrected inference about the extreme Expected Shortfall in the general Max-Domain of Attraction. *Information and Inference: A Journal of the IMA* **14** iaaf023. https://doi.org/10.1093/imaiai/iaaf023

Datta, S., Guha, R., Shudde, R. and Johnson, V. E. (2025a). Bayes factors functions based on test statistics and non-local moment prior densities. *Statistics & Probability Letters* **219** 110330. https://doi.org/10.1016/j.spl.2024.110330

Datta, S., Guha, R., Shudde, R. and Johnson, V. E. (2025b). On Bayes factor functions. *Bayesian Analysis* **20** 1399–1427. https://doi.org/10.1214/25-BA1562

Evans, M. (2016). Measuring statistical evidence using relative belief. *Computational and Structural Biotechnology Journal* **14** 91–96. https://doi.org/10.1016/j.csbj.2015.12.001

Grünwald, P. D. (2024). Beyond Neyman–Pearson: E-values enable hypothesis testing with a data-driven alpha. *Proceedings of the National Academy of Sciences* **121** e2302098121. https://doi.org/10.1073/pnas.2302098121

Halsey, L. G. (2019). The reign of the p-value is over: What alternative analyses could we employ to fill the power vacuum? *Biology Letters* **15** 1–8. https://doi.org/10.1098/rsbl.2019.0174

Held, L. and Ott, M. (2016). How the maximal evidence of p-values against point null hypotheses depends on sample size. *The American Statistician* **70** 335–341. https://doi.org/10.1080/00031305.2016.1209128

Held, L. and Ott, M. (2018). On p-values and Bayes factors. *Annual Review of Statistics and Its Application* **5** 393–419. https://doi.org/10.1146/annurev-statistics-031017-100307

Huisman, L. (2023). Are P-values and Bayes factors valid measures of evidential strength? *Psychonomic Bulletin & Review* **30** 932–941. https://doi.org/10.3758/s13423-022-02205-x

Hwang, T. J., Carpenter, D., Lauffenburger, J. C., Wang, B., Franklin, J. M. and Kesselheim, A. S. (2016). Failure of investigational drugs in late-stage clinical development and publication of trial results. *JAMA Internal Medicine* **176** 1826–1833. http://dx.doi.org/10.1001/jamainternmed.2016.6008

Ioannidis, J. P. A. (2005). Why most published research findings are false. *PLOS Medicine* **2** 696–701. https://doi.org/10.1371/journal.pmed.1004085

JASP (2026). JASP (Version 0.98.0)[Computer software].

Jeffreys, H. (1936). Further significance tests. *Mathematical Proceedings of the Cambridge Philosophical Society* **32** 416–445. https://doi.org/10.1017/S0305004100019125

Johnson, V. E. (2005). Bayes factors based on test statistics. *Journal of the Royal Statistical Society Series B: Statistical Methodology* **67** 689–701. https://doi.org/10.1111/j.1467-9868.2005.00521.x

Johnson, V. E. (2008). Properties of Bayes factors based on test statistics. *Scandinavian Journal of Statistics* **35** 354–368. https://doi.org/10.1111/j.1467-9469.2007.00576.x

Johnson, V. E. (2013). Revised standards for statistical evidence. *Proceedings of the National Academy of Sciences* **110** 19313–19317. https://doi.org/10.1073/pnas.1313476110

Johnson, V. E., Pramanik, S. and Shudde, R. (2023). Bayes factor functions for reporting outcomes of hypothesis tests. *Proceedings of the National Academy of Sciences* **120** e2217331120. https://doi.org/10.1073/pnas.2217331120

Kass, R. E. and Raftery, A. E. (1995). Bayes factors. *Journal of the American Statistical Association* **90** 773–795. https://doi.org/10.1080/01621459.1995.10476572

Kass, R. E. and Wasserman, L. (1995). A reference Bayesian test for nested hypotheses and its relationship to the Schwarz criterion. *Journal of the American Statistical Association* **90** 928–934. https://doi.org/10.2307/2291327

Kooperberg, C. and Stone, C. J. (1992). Logspline density estimation for censored data. *Journal of Computational and Graphical Statistics* **1** 301–328. https://doi.org/10.1080/10618600.1992.10474588

Lee, M. D. and Wagenmakers, E.-J. (2014). *Bayesian cognitive modeling: A practical course*. Cambridge University Press. https://doi.org/10.1017/CBO9781139087759

Ly, A., Verhagen, J. and Wagenmakers, E.-J. (2016). An evaluation of alternative methods for testing hypotheses, from the perspective of Harold Jeffreys. *Journal of Mathematical Psychology* **72** 43–55. https://doi.org/10.1016/j.jmp.2016.01.003

Marsman, M. and Wagenmakers, E.-J. (2017). Three insights from a Bayesian interpretation of the one-sided P value. *Educational and Psychological Measurement* **77** 529–539. https://doi.org/10.1177/0013164416669201

Masson, M. E. J. (2011). A tutorial on a practical Bayesian alternative to null-hypothesis significance testing. *Behavior Research Methods* **43** 679–690. https://doi.org/10.3758/s13428-010-0049-5

Miron, L., Gonçalves, R. S. and Musen, M. A. (2020). Obstacles to the reuse of study metadata in ClinicalTrials.gov. *Scientific Data* **7** 1–14. https://doi.org/10.1038/s41597-020-00780-z

Morey, R. D., Romeijn, J.-W. and Rouder, J. N. (2016). The philosophy of Bayes factors and the quantification of statistical evidence. *Journal of Mathematical Psychology* **72** 6–18. https://doi.org/10.1016/j.jmp.2015.11.001

Morey, R. D. and Rouder, J. N. (2026). BayesFactor: Computation of Bayes factors for common designs. R package version 0.9.12-4.8.

Mulder, J., Wagenmakers, E.-J. and Marsman, M. (2022). A generalization of the Savage–Dickey density ratio for testing equality and order constrained hypotheses. *The American Statistician* **76** 102–109. https://doi.org/10.1080/00031305.2020.1799861

Nathoo, F. S. and Masson, M. E. J. (2016). Bayesian alternatives to null-hypothesis significance testing for repeated-measures designs. *Journal of Mathematical Psychology* **72** 144–157. https://doi.org/10.1016/j.jmp.2015.03.003

Neath, A. A. and Cavanaugh, J. E. (2012). The Bayesian information criterion: Background, derivation, and applications. *WIREs Computational Statistics* **4** 199–203. https://doi.org/10.1002/wics.199

Raftery, A. E. (1995). Bayesian model selection in social research. *Sociological Methodology* **25** 111–163. https://doi.org/10.2307/271063

Rostgaard, K. (2023). Simple nested Bayesian hypothesis testing for meta-analysis, Cox, Poisson and logistic regression models. *Scientific Reports* **13** 1–11. https://doi.org/10.1038/s41598-023-31838-8

Sinharay, S. and Stern, H. S. (2005). An empirical comparison of methods for computing Bayes factors in generalized linear mixed models. *Journal of Computational and Graphical Statistics* **14** 415–435. https://doi.org/10.1198/106186005X47471

Stan (2025). RStan: the R interface to Stan. R package version 2.32.7.

Sun, D., Gao, W., Hu, H. and Zhou, S. (2022). Why 90% of clinical drug development fails and how to improve it? *Acta Pharmaceutica Sinica B* **12** 3049–3062. https://doi.org/10.1016/j.apsb.2022.02.002

Volinsky, C. T. and Raftery, A. E. (2000). Bayesian information criterion for censored survival models. *Biometrics* **56** 256–262. https://doi.org/10.1111/j.0006-341x.2000.00256.x

Wagenmakers, E.-J. (2007). A practical solution to the pervasive problems of p values. *Psychonomic Bulletin & Review* **14** 779–804. https://doi.org/10.3758/bf03194105

Wagenmakers, E.-J. (2022). Approximate objective Bayes factors from p-values and sample size: The $3p\sqrt{n}$ rule. *PsyArXiv* **1** 1–50. https://doi.org/10.31234/osf.io/egydq

Wagenmakers, E.-J., Bartoš, F. and Aust, F. (2026). Approximate Bayes factors are approximate e-processes. *SSRN*. https://doi.org/10.2139/ssrn.6138254

Wagenmakers, E.-J. and Ly, A. (2023). History and nature of the Jeffreys–Lindley paradox. *Archive for History of Exact Sciences* **77** 25–72. https://doi.org/10.1007/s00407-022-00298-3

Wasserstein, R. L. and Lazar, N. A. (2016). The ASA statement on p-values: Context, process, and purpose. *The American Statistician* **70** 129–133. https://dx.doi.org/10.1080/00031305.2016.1154108

Wei, Z., Nathoo, F. S. and Masson, M. E. J. (2023). Investigating the relationship between the Bayes factor and the separation of credible intervals. *Psychonomic Bulletin & Review* **30** 1759–1781. https://doi.org/10.3758/s13423-023-02295-1

Yuan, Y. and Johnson, V. E. (2008). Bayesian hypothesis tests using nonparametric statistics. *Statistica Sinica* **18** 1185–1200.